Data Structures and Algorithms

Prototype Questions with Answers

1. A mathematical expression is in postfix form if the operators appear after their operands. For example, 12+ represents the sum of 1 and 2 and its evaluated result 3. Suppose you are required to write an algorithm to read a postfix mathematical expression and print its evaluated result when the user enters the equality operator (=). For example, your program should print 21 when the user enters 12+34+*=↵.

   a. What type of data structure would be suitable for implementing your algorithm?

      Integer Stack

   b. What assumptions would you make to simplify your algorithm?

      *Assume that the expression contains only single-digit operands (0–9) and the basic arithmetic operators (+, -, *, /).*

      Assume that the postfix expression entered by the user is valid.

   c. Specify your algorithm in pseudocode. You may assume that the required data structure mentioned in 1(a) is available.

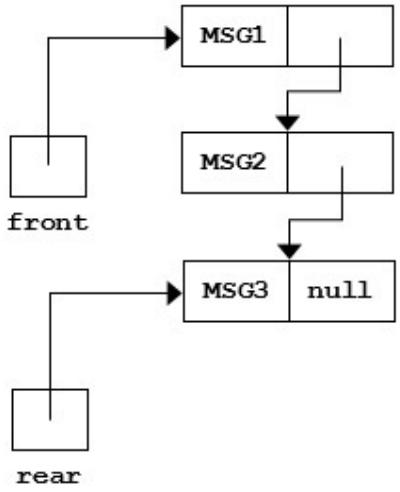
```
ALGORITHM evaluatePostfix()
1   create an empty integer Stack
2   READ character
3   LOOP (character not equals to =)
4      IF (character is an operator)
5         operand2 ← POP from Stack
6         operand1 ← POP from Stack
7         result ← operand1 operator operand2
8         PUSH result onto Stack
9      ELSE
10        convert character to integer
11        push integer into stack
12     END IF
13     READ character
14  END LOOP
15  evaluatedResult ← POP from Stack
16  PRINT evaluatedResult
END evaluatePostfix
```

2. A Short Message Service (SMS) gateway feeds incoming text messages (feed) from different mobile networks at one end of a certain type of data structure and releases them (release) from the other end of the same data structure to their own mobile network for delivery to their intended recipients in the order they arrive.

   a. What type of data structure would be suitable for implementing this SMS gateway? Note that the data structure should be capable of handling any number of incoming text messages.

      Linked List-based Queue

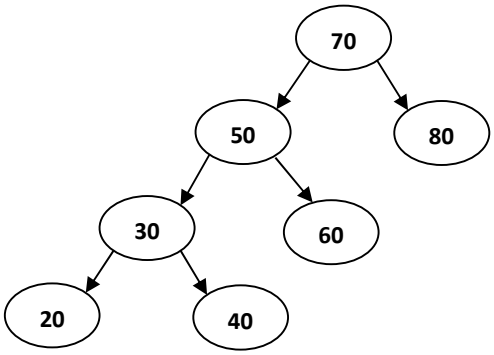
b. If three text messages, namely MSG1, MSG2 and MSG3 have been fed into the data structure mentioned in 2(a), illustrate an overview of the data structure in the system's memory.



3.

a. Store the incoming numbers 70, 50, 30, 80, 60, 40 and 20 as per the following recursive algorithm in a binary tree data structure.

```
insert(number, node)
   if node is empty
      store number in node
   else if (number < value in node)
           insert(number, left child of node)
        else
           insert(number, right child of node)
```



b. What would be the output if you traverse the above binary tree in pre-order and print the contents of nodes?

70, 50, 30, 20, 40, 60, 80

4. Consider the following algorithm specified in pseudo code.

```
ALGORITHM sum(REF a <ARRAY>, VAL n <INTEGER>)
1   IF (n > 0)
2      RETURN sum(a, n-1) + a[n-1]
3   END IF
4   RETURN 0
END sum
```

a. Determine the space complexity of the above algorithm. Show all your workings clearly. Assume that the target computer for this algorithm is a 32-bit machine.

Each function call requires space on environment stack for "a", "n" and also for the return address of calling instruction.
As array is passed by reference, the parameter "a" requires 4 bytes to store its reference on a 32-bit machine.
As "n" is an integer parameter, it also requires 4 bytes.
As return address is a pointer, it requires another 4 bytes on a 32-bit machine.
Recursion depth is n + 1.
Thus, space complexity S(n) = 12(n + 1)

b. Determine the time complexity of the above algorithm using the Step Count approach. Show all your workings clearly.

```
ALGORITHM sum(REF a <ARRAY>, VAL n <INTEGER>)
1   IF (n > 0)    -------------------------     (n+1) steps
2      RETURN sum(a, n-1) + a[n-1]  -------     n steps
3   END IF
4   RETURN 0      -------------------------     1
END sum
```

Total number of steps = (n + 1) + n + 1 = 2n +
2 Thus, time complexity T(n) = 2n + 2